5.5.3 Equation of a Straight line parallel to the Y axis

Let CD be a straight line parallel to Y axis, which is at a distance 'c'. Then x coordinate of every point on CD is 'c'. The equation of CD is $x = c$. (fig 5.30)

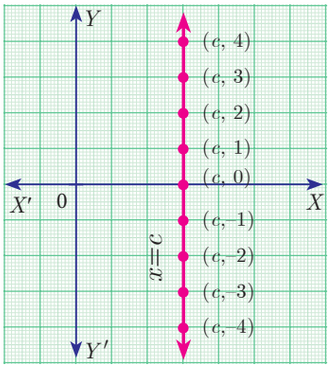


Fig. 5.30

Note

- If $c > 0$, then the line $x=c$ lies right to the side of the Y axis
- If $c < 0$, then the line $x=c$ lies left to the side of the Y axis
- If $c = 0$, then the line $x=c$ is the Y axis itself.

Example 5.17 Find the equation of a straight line passing through $(5,7)$ and is (i) parallel to X axis (ii) parallel to Y axis.

Solution (i) The equation of any straight line parallel to X axis is $y=b$.

Since it passes through $(5,7)$, $b = 7$.

Therefore, the required equation of the line is $y=7$.

(ii) The equation of any straight line parallel to Y axis is $x=c$

Since it passes through $(5,7)$, $c = 5$

Therefore, the required equation of the line is $x=5$.

5.5.4 Slope-Intercept Form

Every straight line that is not vertical will cut the Y axis at a single point. The y coordinate of this point is called y intercept of the line.

A line with slope m and y intercept c can be expressed through the equation $y=mx+c$

We call this equation as the slope-intercept form of the equation of a line.

DO YOU KNOW?

- If a line with slope m, $m\neq0$ makes x intercept d, then the equation of the straight line is $y = m(x-d)$.
- $y = mx$ represent equation of a straight line with slope m and passing through the origin.

Example 5.18 Find the equation of a straight line whose

(i) Slope is 5 and y intercept is -9 (ii) Inclination is $45°$ and y intercept is 11

Solution (i) Given, Slope $= 5$, y intercept, $c = -9$

Therefore, equation of a straight line is $y = mx + c$

$$y = 5x - 9 \Rightarrow 5x - y - 9 = 0$$

(ii) Given, $\theta = 45°$, y intercept, $c = 11$

Slope $m = \tan\theta = \tan 45° = 1$

Therefore, equation of a straight line is of the form $y = mx + c$

Hence we get, $y = x + 11 \Rightarrow x - y + 11 = 0$

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